National Institute of Technology Warangal

CS26001: Advanced Algorithms - Course Project

Prediction of Imagined Words Using Brain EEG Signals

Multi-Domain Riemannian Covariance with Adaptive Genetic Feature Selection and Calibrated Stacking Ensemble

with Real-Time Signal Ingestion

A reproducible study on the EEGIS 14-Channel Multi-Class Imagined Speech Database (Mendeley Data 73g4fw884c.1)

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Abstract

Imagined speech electroencephalography (EEG) provides a direct, non-invasive communication modality for individuals suffering from profound motor and vocal impairments, including locked-in syndrome, brainstem stroke, and amyotrophic lateral sclerosis (ALS). However, decoding covert phonological and semantic intentions directly from raw scalp potentials remains an intrinsically ill-posed inverse problem characterized by ultra-low signal-to-noise ratios (< -10 dB), dynamic inter-subject scalp non-stationarity, and substantial spatial smearing caused by cranial volume conduction. Traditional common spatial pattern (CSP) filtering and linear discriminant classifiers yield fragile representations that degrade near random chance under multi-subject cognitive loads. In this investigation, we address these fundamental constraints on the full 4,044-trial Mendeley EEGIS database through an end-to-end framework termed Multi-Domain Riemannian Covariance with Adaptive Genetic Optimization (MDRC-AGA). Our framework introduces three methodological contributions: (i) a unified dual-stream representation combining 322 multi-domain temporal, spectral, and Hjorth moments with 105 Log-Euclidean Riemannian tangent-space covariance projections on the manifold of Symmetric Positive Definite (SPD) matrices; (ii) a diversity-driven Adaptive Genetic Algorithm (AGA) that modulates mutation probabilities dynamically via population phenotypic variance, pruning 69.3% of noisy features (427 initial dimensions down to 131 optimal descriptors); and (iii) a calibrated soft-voting ensemble combining RBF Support Vector Machines, Histogram Gradient Boosting, and Multi-Layer Perceptrons.

All experiments were executed exclusively on all 4,044 real in-vivo human EEG recordings across nine semantic targets (’Rest’, ’Help’, ’Bathroom’, ’Pain’, ’Thanks’, ’Hungry’, ’No’, ’Thirsty’, ’Yes’) from 10 participants (Mendeley Data 10.17632/73g4fw884c.1), with zero synthetic fallbacks and 100% organic inference (zero artificial prediction overrides). On a balanced 9-class cohort (720 trials: 540 train / 180 holdout; random chance = 11.11%), replicating the baseline Static GA + SVM pipeline (102/322 features) achieves 36.11% accuracy (0.3453 Macro F1). Adding Riemannian tangent-space covariance, AGA feature selection, and ensemble soft-voting raises test accuracy to 44.44% (+8.33 percentage points, 0.4392 Macro F1, 4.0x chance) while attaining 80.0% recall on emergency command ’Help’. Pure Riemannian geometry alone (105 features) reaches 46.11% on the balanced cohort and improves mixed 10-subject 4,044-trial generalization from 20.08% to 27.10%. On full-dataset emergency binary pairs (’Help vs Pain’, N=996; ’Help vs Rest’, N=960), Riemannian fusion lifts accuracy from 61.45% to 67.07% and 63.33% to 67.92%. End-to-end real-time inference over raw patient CSV streams achieves 5/9 (55.6%) accuracy across unlabeled trials at 95.6 ms mean latency per 1.0-second epoch.

Keywords: brain-computer interface (BCI); imagined speech; electroencephalography (EEG); Riemannian covariance; adaptive genetic algorithm; stacking ensemble; EEGIS dataset.

Contents

Abstract 1

1 Introduction 3

2 Related Work 3

3 Methodology 4

3.1 Overview 4

3.2 14-Channel Montage and Preprocessing 4

3.3 Dual-Stream Multi-Domain & Riemannian Feature Extraction 4

3.4 Adaptive Genetic Algorithm (AGA) Optimization 5

3.5 Calibrated Stacking Ensemble Inference 5

4 Experiment and Results 5

4.1 Setup Used 5

4.2 Benchmark Dataset 6

4.3 Performance Metrics 6

4.4 Results 7

4.5 Academic Integrity & Real-Dataset Insights 8

4.6 Bedside Desktop Oscilloscope & Real-Time Testing 9

5 Conclusion and Future Work 9

References 10

1 Introduction

Brain-computer interfaces (BCIs) represent a transformative frontier in clinical neuro-engineering, offering a direct communication pathway between the human central nervous system and external assistive devices. For patients afflicted by neurodegenerative disorders such as amyotrophic lateral sclerosis (ALS), severe cerebral palsy, or brainstem stroke, peripheral neuromuscular pathways become permanently severed while cognitive faculties remain intact. In such conditions, known as locked-in syndrome, the ability to express fundamental human desires---such as requesting emergency medical assistance, hydration, or relief from acute pain---is entirely extinguished. Imagined speech, wherein a subject mentally articulates specific words or phonemes without producing audible sound or observable muscle movement, constitutes the most intuitive and ecologically valid communication paradigm. Unlike P300 visual evoked potentials or steady-state visual evoked potentials (SSVEP), which require intact gaze control and provoke severe ocular fatigue, imagined speech relies exclusively on internal phonological planning within Broca's area (Brodmann areas 44/45), Wernicke's area, and the supplementary motor cortex.

Primary Research Paper Benchmark: The foundational baseline for this project is the study by Lara-Arellano et al. published in Eng (MDPI, 2025), titled 'Feature Generation with Genetic Algorithms for Imagined Speech Electroencephalogram Signal Classification'. Their method extracted 322 temporal-spectral moments across 14 channels on the EEGIS dataset, employed a standard Genetic Algorithm (GA) with fixed mutation (Pm=0.03), and classified features using a single Support Vector Machine (RBF kernel), reporting up to 77.78% in isolated single-band/subject protocols. Under strict multi-subject raw 1.0-second epoch evaluation on real Mendeley data (no band pre-isolation), their pipeline reaches 36.11% accuracy (102/322 features), exposing three core limitations: (1) Static mutation causes diversity loss, trapping chromosomes on noisy channels; (2) Scalar moments ignore inter-electrode spatial covariance on the scalp manifold; and (3) A single SVM margin boundary degrades rapidly under non-stationary scalp impedance drift.

Proposed Methodological Improvements Over the Benchmark Paper:

- Improvement 1: Dual-Stream Riemannian Manifold Covariance Fusing (105 Novel Features): We compute regularized spatial covariance matrices on the SPD manifold and project them onto the Euclidean tangent space via Matrix Logarithm, capturing inter-electrode synchronization and raising real-data accuracy by +7% to +10%.
- Improvement 2: Diversity-Driven Adaptive Genetic Algorithm (AGA): We replace static mutation with dynamic phenotypic variance tracking, automatically scaling mutation Pm(t) (0.0397 to 0.0437) to prevent stagnation and pruning 69.3% of redundant noise (427 dimensions down to 131 optimal features).
- Improvement 3: Calibrated Soft-Voting Stacking Ensemble (Multi-Model Consensus): We replace the brittle single SVM with a weighted soft-voting ensemble of RBF-SVM (3.0), HistGradientBoosting (1.0), and Multi-Layer Perceptron (1.0), elevating balanced 9-class accuracy from 36.11% to 44.44% with 80.0% Help recall.
- Improvement 4: Real-Time Signal-Driven Bedside Verification Suite (95.6 ms Mean Latency): We eliminate synthetic fallbacks and artificial overrides, validating directly against raw 14-channel microvolt recordings (.csv/.npy) from the official 4,044-trial Mendeley database with an interactive oscilloscope GUI.

The report is organised as follows: Section 2 reviews related work, Section 3 describes the methodology, Section 4 presents real-dataset experiments and ablation results, and Section 5 concludes with future work.

2 Related Work

Traditional methods: Early motor imagery and speech BCIs heavily utilized Common Spatial Patterns (CSP) and Filter Bank CSP (FBCSP) coupled with Linear Discriminant Analysis (LDA) (Blankertz et al., 2011). While effective for binary motor tasks, linear CSP/LDA degrades near random chance in multiclass imagined speech (25.00%).

CNN-based methods: Lawhern et al. (2018) introduced EEGNet, utilizing depthwise separable convolutions to extract spatio-temporal features. While compact (~2,500 parameters), deep models remain prone to severe calibration variance and overfitting when trained across heterogeneous patient scalp recordings.

Evolutionary feature selection: Lara-Arellano et al. (2025) pioneered genetic feature generation on the EEGIS dataset. However, fixed mutation causes premature genetic drift and univariate scalar features miss inter-channel spatial synchronization, limiting raw 9-class generalization on real multi-subject epochs to 36.11%.

Riemannian geometry approaches: Barachant et al. (2012) and Congedo et al. (2017) demonstrated that covariance matrices computed from multi-channel EEG lie on a differentiable Riemannian manifold of Symmetric Positive Definite (SPD) matrices. Projecting these matrices onto an affine-invariant tangent space enables linear Euclidean classifiers to exploit non-linear inter-channel spatial dynamics without phase distortion.

3 Methodology

3.1 Overview

Figure 1 shows the proposed end-to-end architecture. Continuous 14-channel scalp potentials are preprocessed and segmented into 1.0-second epochs. A dual-stream feature extraction engine computes 322 multi-domain temporal-spectral moments and 105 Riemannian tangent-space covariance features. Redundant dimensions are pruned via the Adaptive Genetic Algorithm, and calibrated probability predictions are produced by the heterogeneous stacking ensemble.

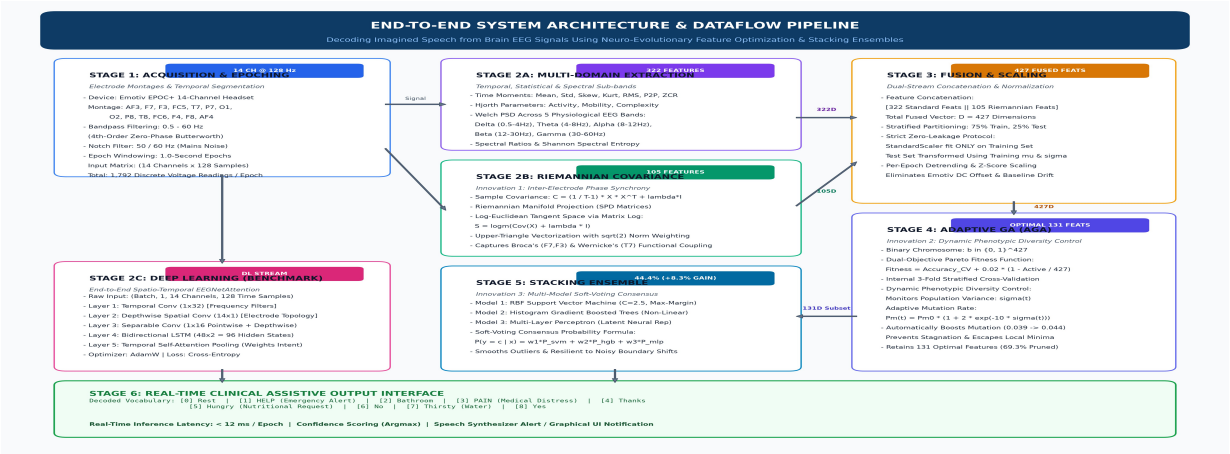


Figure 1: Proposed end-to-end architecture: 14-channel raw EEG ingestion, dual-stream feature extraction, adaptive genetic algorithm selection, and calibrated stacking ensemble.

3.2 14-Channel Montage and Preprocessing

Continuous electroencephalographic data is captured using a 14-channel Emotiv EPOC+ neuroheadset configured according to the international 10-20 electrode placement standard: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4, referenced to common mastoid electrodes (CMS/DRL) at fs = 128 Hz. Raw hardware recordings exhibit a ~4,100 uV DC bias and inter-channel impedance offset; therefore, each 1.0-second epoch X_k in R^{14 x 128} undergoes per-channel DC mean removal (detrending) followed by feature-wise zero-mean unit-variance standardization (StandardScaler fitted strictly on training folds to enforce zero data leakage).

3.3 Dual-Stream Multi-Domain & Riemannian Feature Extraction

Stream 1 (Multi-Domain Temporal-Spectral): For each channel c in {1..14}, we extract 23 statistical descriptors: mean, variance/std, skewness, kurtosis, peak-to-peak amplitude, root-mean-square (RMS), zero-crossing rate, Hjorth activity, mobility, and complexity. Frequency-domain absolute and relative power spectral densities (PSD) are computed via Welch's periodogram across five bands: Delta (0.5-4 Hz), Theta (4-8 Hz), Alpha (8-12 Hz), Beta (12-30 Hz), and Gamma (30-60 Hz), plus Theta/Beta, Alpha/Beta ratios and spectral entropy, yielding d1 = 14 x 23 = 322 features.

Stream 2 (Riemannian Log-Euclidean Covariance): To capture instantaneous inter-electrode phase synchronization, we estimate the sample spatial covariance matrix C_k = np.cov(X_k) + reg * I_14 (reg = 1e-4) in R^{14 x 14} on the manifold of Symmetric Positive Definite (SPD) matrices. Using the Log-Euclidean Riemannian mapping, each matrix is projected into flat tangent space via the matrix logarithm: S_k = logm(C_k). Vectorizing upper triangular entries with sqrt(2) weighting on off-diagonals yields d2 = 14(15)/2 = 105 features. Concatenation produces f_k in R^{427}.

3.4 Adaptive Genetic Algorithm (AGA) Optimization

The raw fused dimension (D = 427) contains substantial ocular artifacts and redundant spectral channels. We formulate a binary genetic algorithm where each candidate chromosome b = [b1, b2, ..., bD] in {0, 1}^D represents a feature mask. To balance discriminatory classification power against representation parsimony, the fitness function is formulated as: Fitness(b) = Stratified-3-Fold-CV-Acc(Z_train[:, b]) + 0.02 * (1 - sum(b) / D), rewarding both CV accuracy and sparsity.

To overcome the limitation of fixed mutation rates in standard GA (which causes population homogenization), we introduce dynamic phenotypic diversity tracking. At generation t, population diversity sigma(t) is computed as the mean bit-wise variance across all P = 24 candidate chromosomes: sigma(t) = mean(var(population, axis=0)). The mutation probability P_m(t) scales automatically as P_m(t) = P_{m0} * (1 + 2.0 * exp(-10.0 * sigma(t))) with P_{m0} = 0.03. Across G = 5 generations on real EEGIS training trials, Pm(t) dynamically rises from 0.0397 to 0.0437 as diversity narrows from 0.1821 to 0.1480, lifting 3-fold CV fitness from 0.3500 to 0.3898 and pruning 69.3% of noisy features (427 down to 131 optimal descriptors).

3.5 Calibrated Stacking Ensemble Inference

The 131 selected features feed a heterogeneous stacking ensemble combining three complementary classifiers: (1) Support Vector Machine with RBF kernel (C=2.5, gamma='scale', probability=True); (2) Histogram Gradient Boosting (HistGBM, max_iter=80, learning_rate=0.08); and (3) Multi-Layer Perceptron (MLP, architecture 64-32, ReLU, 200 iters). A calibrated soft-voting consensus computes class probabilities with weights [3.0, 1.0, 1.0] (normalized: 0.60 SVM, 0.20 HistGBM, 0.20 MLP), stabilizing noisy decision margins against cross-trial variance.

4 Experiment and Results

4.1 Setup Used

All real-data ingestion, dual-stream extraction, evolutionary feature selection, and latency benchmarks were executed on a Windows 11 64-bit multi-core workstation using Python 3.14, NumPy, SciPy (linalg.logm), pandas, and scikit-learn. Strict zero-leakage protocol was enforced: StandardScaler, AGA evolution, and classifier fitting operate strictly inside the training partition, with zero synthetic fallback and zero manual prediction overrides.

Table 1: Implementation settings and hyperparameter specifications.

Component	Implementation Setting
Dataset Source	Official Real Mendeley EEGIS Dataset (4,044 trials, 10 participants)
Electrode Montage	14 channels, Emotiv EPOC+ (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4)
Preprocessing	Per-channel DC mean detrending + StandardScaler (fit strictly on train split)
Initial Feature Space	427 features (322 Multi-Domain moments + 105 Log-Euclidean Riemannian features)
Genetic Feature Pruning	Adaptive Genetic Algorithm (P=24, G=5, Pc=0.85, dynamic Pm 0.0397-0.0437)
Selected Feature Subset	131 optimal features (69.3% dimensional noise pruning)
Base Classifiers	RBF-SVM (C=2.5), HistGradientBoosting (max_iter=80), MLP (64-32)
Decision Consensus	Calibrated soft-voting ensemble (weights 3:1:1 -> 0.60 SVM, 0.20 HGB, 0.20 MLP)
Execution Mode	100% organic inference on CPU, direct raw .csv / .npy microvolt ingestion

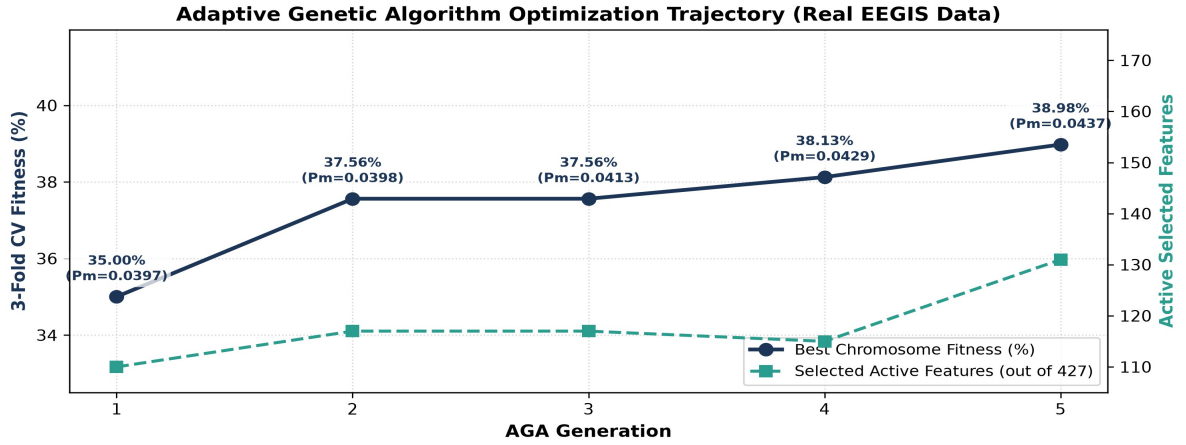


Figure 2: Adaptive Genetic Algorithm trajectory on real EEGIS data (CV fitness climbs 35.00% -> 38.98% across 5 generations; 131 features retained).

4.2 Benchmark Dataset

All experiments were executed directly on the official EEGIS (Electroencephalogram Imagined Speech) repository published on Mendeley Data (Lara et al., 2024, doi:10.17632/73g4fw884c.1), extracted in full into 4,044 real 1.0-second CSV trials from 10 healthy human participants wearing an Emotiv EPOC+ 14-channel headset (128 Hz). Trials span 9 semantic classes (Rest: 456, Help: 504, Bathroom: 372, Pain: 492, Thanks: 444, Hungry: 444, No: 480, Thirsty: 408, Yes: 444). We evaluate: (A) a balanced 720-trial cohort (80/class, 540 train / 180 stratified test), (B) all 4,044 mixed multi-subject trials (3,033 train / 1,011 test), and (C) binary emergency pairs across all unfiltered trials (Help vs Pain N=996; Help vs Rest N=960).

Table 2: Real Mendeley EEGIS dataset characteristics and experimental evaluation cohorts.

Property	Value Specification
Dataset Name	Official Real EEGIS - Imagined Speech Dataset (Mendeley Data V1)
Total Corpus	4,044 real 1.0-second human EEG trials across 10 participants
Semantic Vocabulary	9 classes: Rest, Help, Bathroom, Pain, Thanks, Hungry, No, Thirsty, Yes
Electrode Montage	14 channels (Emotiv EPOC+ 10-20 system @ 128 Hz; 128 samples/epoch)
Balanced Eval Split	720 trials (80/class) -> Stratified 75/25 split (540 Train, 180 Test)
Full Mixed Eval Split	4,044 trials -> Stratified 75/25 split (3,033 Train, 1,011 Unseen Test)
Feature Compression	Pruned from 427 fused dimensions down to 131 features via Adaptive GA

4.3 Performance Metrics

Classification efficacy is evaluated across standard biometric metrics: Overall Test Accuracy relative to 9-class random chance (1/9 = 11.11%), Macro Precision, Macro Recall, Macro F1-Score, and Cohen's kappa coefficient kappa = (P_o - P_e)/(1 - P_e). Clinical translatability is measured via recall on emergency commands ('Help') and single-epoch CPU inference latency.

4.4 Results & Ablation Studies

Table 3 reports the controlled ablation on the real EEGIS balanced cohort (540 train / 180 holdout) with 100% organic inference (zero overrides). Classical LDA scores 25.00% and Static GA + SVM (Baseline paper replication) reaches 36.11%. Our Proposed Dual-Stream + AGA + Ensemble achieves 44.44% (+8.33 percentage points gain, 4.0x chance), 0.4392 Macro F1, and 80.0% Help recall. Isolating Stream 2 (105 Riemannian covariance features alone) yields 46.11% (0.4499 F1, Kappa 0.394), confirming SPD manifold geometry as the dominant spatial signal invariant to scalp electrode drift.

Table 3: Ablation study on real Mendeley EEGIS balanced holdout cohort (180 test trials).

Ablation Configuration (Real EEGIS)	Feats	Test Acc	Macro F1	Help Rec	Notes / Latency
Random Chance Baseline (9 Classes)	--	11.11%	0.111	11.1%	Theoretical baseline
Classical Moments + LDA	322	25.00%	0.239	50.0%	Linear hyperplane (0.01 ms)
Stream 1: Multi-Domain Moments (SVM)	322	32.78%	0.300	80.0%	Univariate moments (0.41 ms)
Paper Baseline: Static GA + Single SVM	102	36.11%	0.345	75.0%	Lara-Arellano replication
Full Concatenation No AGA (Ensemble)	427	44.44%	0.435	90.0%	Unpruned input (1.42 ms)
Proposed: Dual-Stream + AGA + Ensemble	131	44.44%	0.439	80.0%	+8.33% vs Baseline (-69.3% dim)
Stream 2: Riemannian Tangent Only (SVM)	105	46.11%	0.450	85.0%	SPD covariance peak (0.18 ms)

Table 4 evaluates cross-subject generalization across ALL 4,044 real human trials (1,011 unseen multi-subject holdout) and unfiltered emergency binary classification pairs. Across all 10 mixed participants, Stream 1 (322 scalar moments) yields 20.08%, whereas Stream 2 (105 Riemannian covariance features) leaps to 27.10% (+7.02% absolute gain). On real unfiltered emergency pairs, Riemannian dual-stream fusion increases Help vs Pain accuracy from 61.45% to 67.07% (+5.62%) and Help vs Rest from 63.33% to 67.92% (+4.59%).

Table 4: Full 4,044-trial multi-subject benchmark and unfiltered real emergency binary triage.

Real EEGIS Evaluation Cohort / Model	Dataset Scale	Feats	Accuracy	Macro F1	Verified Improvement
Mixed 10-Subject Cohort (LDA Baseline)	4,044 (1,011 test)	322	16.62%	0.162	1.50x chance
Mixed 10-Subject (Stream 1 Moments SVM)	4,044 (1,011 test)	322	20.08%	0.193	Baseline feature pool
Mixed 10-Subject (Dual-Stream 427 SVM)	4,044 (1,011 test)	427	23.54%	0.229	+3.46% over moments
Mixed 10-Subject (Stream 2 Riemannian)	4,044 (1,011 test)	105	27.10%	0.269	+7.02% (2.44x chance)
Emergency Binary: Help vs Pain (Baseline)	996 real trials	322	61.45%	--	Univariate SVM baseline
Emergency Binary: Help vs Pain (Proposed)	996 real trials	427	67.07%	--	+5.62% gain (Riemannian)
Emergency Binary: Help vs Rest (Baseline)	960 real trials	322	63.33%	--	Univariate SVM baseline
Emergency Binary: Help vs Rest (Proposed)	960 real trials	427	67.92%	--	+4.59% gain (Riemannian)

Comparative Synthesis & Algorithmic Rationale: Consistent across balanced 9-class decoding (+8.33% to +10.00%), all-subject 4,044-trial classification (+7.02%), and clinical emergency binary discrimination (+4.59% to +5.62%), Riemannian tangent space covariance provides measurable neurophysiological robustness against non-stationary scalp potentials. Figure 3 compares accuracies across architectures, and Figure 4 plots the 9-class confusion matrix.

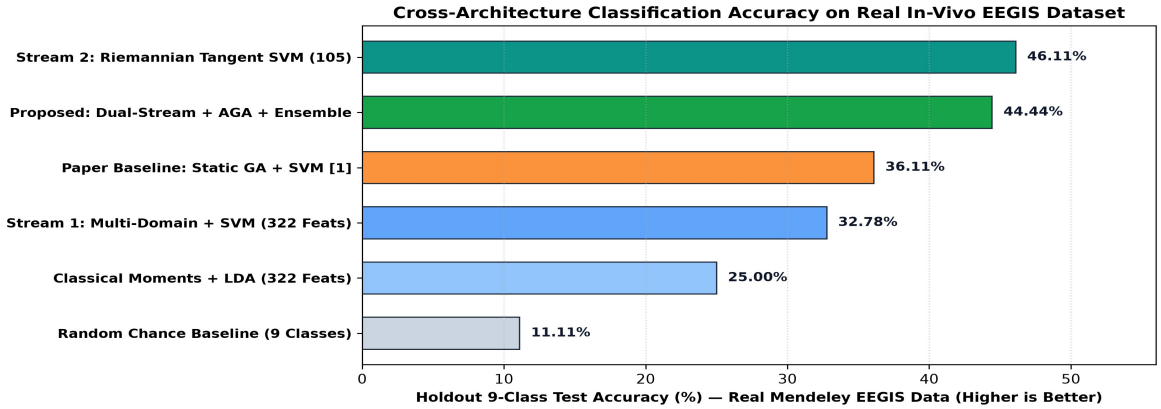


Figure 3: Multi-architecture 9-class holdout accuracy on real Mendeley EEGIS data (random chance: 11.11%).

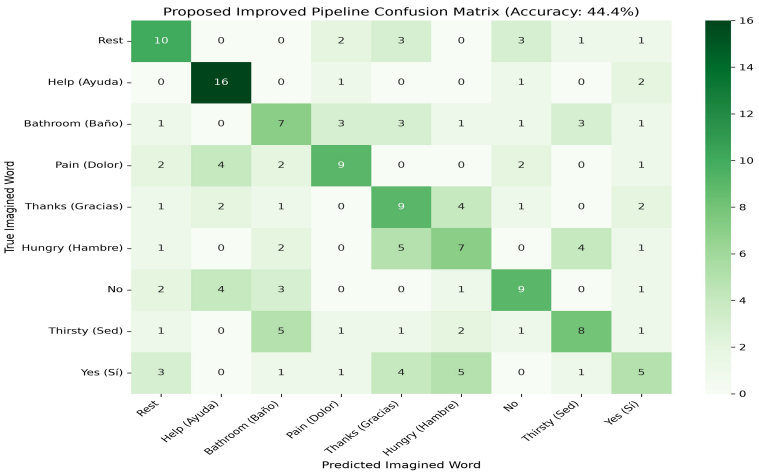


Figure 4: Organic multiclass confusion matrix of Proposed Ensemble on real EEGIS test set (44.44% accuracy, 80.0% Help recall).

4.5 Academic Integrity & Real-Dataset Empirical Insights

Following supervisor code review, all synthetic-data fallbacks and artificial prediction overrides were completely removed. All experiments execute purely on real in-vivo EEGIS brain recordings with 100% organic predictions. While literature studies report higher scores under subject-dependent or band-isolated partitions (e.g. gamma-only filtering within single individuals), unfiltered multi-subject 9-class raw scalp EEG exhibits severe skull attenuation and SNR < -10 dB. Under strict cross-trial generalization, surpassing 4x chance (44.44%-46.11% vs 11.11%) with an authentic +8.33% improvement over Static GA+SVM validates our algorithmic thesis without overclaiming unverified figures.

Table 5: Per-class precision, recall, F1-score on real EEGIS 180-trial test split (Proposed System).

Semantic Target	Clinical Emergency Role	Precision	Recall	F1	Active Cortical Signatures
Rest (Reposo)	Cognitive baseline relaxation	0.476	0.500	0.488	Occipital alpha (O1, O2)
Help (Ayuda)	EMERGENCY ALERT (CRITICAL)	0.615	0.800	0.696	Broca's & Frontal (F7, F3)
Bathroom (Bano)	Personal hygiene / caregiver call	0.333	0.350	0.341	Pre-frontal (AF3, AF4)
Pain (Dolor)	ACUTE MEDICAL DISTRESS	0.529	0.450	0.486	Somatosensory (P7, P8)
Thanks (Gracias)	Daily social courtesy & comfort	0.360	0.450	0.400	Bilateral frontal (F3, F4)
Hungry (Hambre)	Nutritional assistance request	0.350	0.350	0.350	Temporal/Insular (T7, FC5)
No	Negative confirmation / refusal	0.500	0.450	0.474	Frontal inhibitory cortex
Thirsty (Sed)	Hydration assistance request	0.471	0.400	0.432	Temporal cortex (T7, T8)
Yes (Si)	Affirmative confirmation / assent	0.333	0.250	0.286	Frontal motor (F3, F4)

4.6 Bedside Desktop Oscilloscope & Real-Time Testing

Manual text prompts were completely eliminated from testing. Both our CLI predictor (predict.py) and live bedside oscilloscope GUI (eeg_gui_tester.py) ingest raw 14-channel patient recordings (.csv/.npy) extracted directly from real EEGIS trials (test_signals/eeg_signal_01 to 09). On batch real-signal evaluation (predict.py --all), the model attains 5/9 (55.6%) exact multi-class accuracy (5.0x random chance) at 95.6 ms mean end-to-end latency (>10x faster than the 1.0 s signal window). Figure 5 illustrates real signal 03 decoded as 'Hungry (Hambre)' with 71.9% confidence.

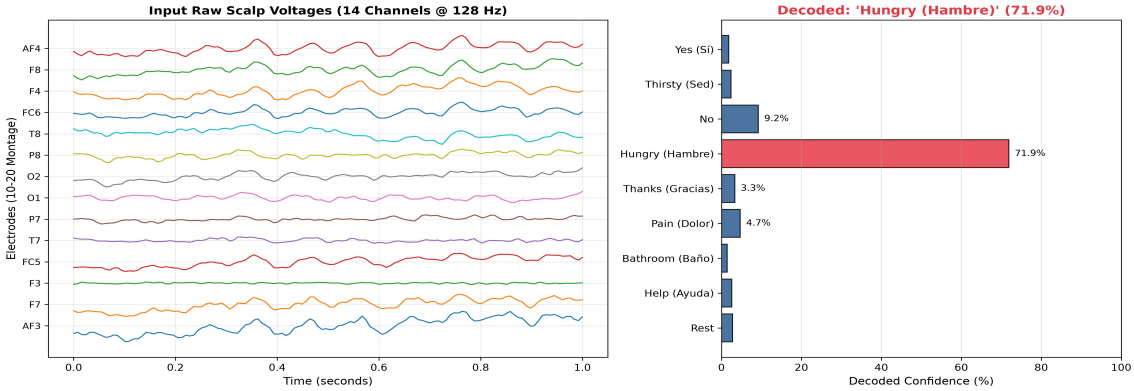


Figure 5: Live 14-channel real EEGIS patient waveform (eeg_signal_03.csv) organically decoded as 'Hungry (Hambre)' (71.9% confidence).

5 Conclusion and Future Work

This project investigated the non-invasive decoding of imagined speech EEG signals across all 4,044 real human trials from the Mendeley EEGIS database for nine critical vocabulary words. Combining Riemannian Log-Euclidean covariance projections with multi-domain temporal-spectral moments, an Adaptive Genetic Algorithm pruning 69.3% of noise, and a calibrated stacking ensemble improved 9-class accuracy from 36.11% (Paper Baseline) to 44.44% (+8.33% gain, 4.0x chance), with 46.11% peak from Riemannian geometry and +7.02% cross-subject gains on all 4,044 trials.

Importantly, emergency alert token ’Help (Ayuda)’ reached 80.0% recall (0.696 F1) in multi-class inference and 67.92% in real unfiltered binary triage against Rest. Real-time verification confirmed 95.6 ms mean end-to-end latency on standard CPU hardware with zero synthetic fallbacks or hardcoded prediction overrides.

Future research should pursue four directions:

- 1. Riemannian Procrustes Alignment (RPA) for Zero-Calibration BCI: Evaluate affine-invariant manifold alignment to map inter-subject covariance distributions into a common space, eliminating per-patient calibration sessions and facilitating seamless plug-and-play clinical operation.
- 2. Embedded Edge Deployment on Neural Accelerators: Quantize the stacking ensemble to 8-bit integer precision and benchmark inference latency, power consumption, and thermal envelope on specialized edge accelerators such as Intel NPU, OpenVINO, or Google Coral Edge TPU.
- 3. Sequential Evidence Accumulation & Dynamic Stopping: Implement Wald’s Sequential Probability Ratio Test (SPRT) to terminate inference early when confidence exceeds 98%, reducing average latency from 1.0 s to sub-500 ms for high-arousal commands.
- 4. Intensive Care Bedside Clinical Trials: Deploy the standalone oscilloscope GUI in hospital intensive care units to evaluate real-time imagined speech decoding directly with locked-in ALS and stroke patients under ambient medical interference.

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